

Causal inference for marketing

a PyMC-ecosystem cookbook

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One marketing business question per technique, the library that fits it, a runnable skeleton, **the statistical detail you need to defend it**, and the decision it drives. Framing follows Michele’s steer: slides lead with the *business decision* and let uncertainty appear as “how confident are we, does it change what we’d do.” This document, by contrast, keeps the math explicit — you won’t expose all of it, but you should own it.

Caveats. Libraries move fast; treat snippets as faithful skeletons to run against your installed versions. Sanity-check the **pymc-bart out-of-sample prediction** pattern (§1) first — the counterfactual-scoring mechanism has shifted across releases.

Statistical foundations (shared notation)

Potential outcomes (Neyman-Rubin). For unit i and binary treatment $T_i \in \{0, 1\}$, define potential outcomes $Y_i(1), Y_i(0)$. We observe only

$$Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0),$$

so the individual effect $Y_i(1) - Y_i(0)$ is **never observed** (the fundamental problem of causal inference — it is a missing-data problem, which is the frame that will feel natural to you). Everything downstream is about imputing the missing counterfactual honestly.

Estimands.

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)], \quad \text{ATT} = \mathbb{E}[Y(1) - Y(0) \mid T = 1], \quad \underbrace{\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]}_{\text{CATE — the HTE object}}.$$

Uplift/personalization lives in $\tau(x)$; program evaluation lives in ATE/ATT.

Identification assumptions (what turns an *estimand* into something estimable from data):

1. **Unconfoundedness / ignorability:** $\{Y(1), Y(0)\} \perp T \mid X$. Given X , treatment is as-good-as-random. Holds *by design* under randomization; an *assumption* otherwise.
2. **Positivity / overlap:** $0 < e(x) < 1$ for all x , where $e(x) = P(T = 1 \mid X = x)$ is the **propensity score**. You need both treated and control units at every x you extrapolate to.
3. **SUTVA:** no interference between units + consistency ($Y = Y(T)$, one version of treatment).

The identification result (backdoor adjustment / g-formula). Under 1-3,

$$\mathbb{E}[Y(t)] = \mathbb{E}_X[\mathbb{E}[Y \mid T = t, X]] = \int \mathbb{E}[Y \mid T = t, X = x] dP(x),$$

so $\tau(x) = \mu_1(x) - \mu_0(x)$ with $\mu_t(x) = \mathbb{E}[Y \mid T = t, X = x]$, and $\text{ATE} = \mathbb{E}_X[\mu_1(X) - \mu_0(X)]$.

The Bayesian thread that unifies every section. Put a model with parameters θ on the response surface $\mu_t(x; \theta)$ (or on the full SCM). The estimand is a **functional of θ** , so the posterior $p(\theta \mid \text{data})$ induces a posterior on the estimand:

$$p(\tau(x) \mid \text{data}) = \int \delta(\tau(x) - [\mu_1(x; \theta) - \mu_0(x; \theta)]) p(\theta \mid \text{data}) d\theta.$$

That is why “credible interval on the CATE” comes for free — no bootstrap, no delta method — and why the do-operator is just posterior-predictive-under-intervention. **Identification is an assumption about the world (untestable); estimation is where Bayes helps.** Keep them separate: priors never rescue a broken identification strategy.

1. Who to target — customer-level uplift • pymc-bart

Business question. Send the €10 discount only to *persuadables*, not sure-things or lost-causes. We need per-customer incremental effect + an honest read on whom we know about.

The math. - Estimand: $\tau(x) = \mu_1(x) - \mu_0(x)$. **Meta-learner choices:** *S-learner* fits one surface $\mu(x, t) = f(x, t)$ and differences $f(x, 1) - f(x, 0)$; *T-learner* fits μ_1, μ_0 separately; *X-learner* imputes unit-level effects then regresses them (better under treatment imbalance). - **BART response surface** (Chipman-George-McCulloch): a sum of m regularized trees

$$f(x) = \sum_{j=1}^m g(x; \mathcal{T}_j, \mathcal{M}_j), \quad Y \mid x \sim \mathcal{N}(f(x), \sigma^2).$$

The **regularization prior** keeps each tree a weak learner: split probability at depth d is $\alpha(1+d)^{-\beta}$ (typically $\alpha=0.95, \beta=2$, so deep trees are unlikely); leaf means $\mu_\ell \sim \mathcal{N}(0, \sigma_\mu^2)$ with $\sigma_\mu \propto 1/\sqrt{m}$ so the *sum* has sensible scale; σ^2 gets an inverse-gamma. Inference is **Bayesian backfitting**: each tree is updated against the partial residual $Y - \sum_{k \neq j} g(x; \mathcal{T}_k, \mathcal{M}_k)$ (pymc-bart uses a particle-Gibbs / PGBART sampler). The posterior over f gives a posterior over $\tau(x_i)$ per customer directly. - **Randomized vs observational.** With a randomized past send, $e(x)$ is constant and unconfoundedness holds by design $\rightarrow$ a plain S/T-learner is unbiased. **Observationally, beware regularization-induced confounding (RIC):** under *targeted selection* (treatment assigned on expected outcome), shrinkage on the treatment contribution biases τ . Fix = **Bayesian Causal Forest** (Hahn-Murray-Carvalho 2020):

$$Y_i = \underbrace{\mu(x_i, \hat{e}(x_i))}_{\text{prognostic BART}} + \underbrace{\tau(x_i)}_{\text{treatment BART}} \quad z_i + \varepsilon_i,$$

with (i) the estimated propensity $\hat{e}(x)$ fed into the prognostic part to absorb the selection, and (ii) a **separate, tighter prior on τ** (fewer trees, smaller leaf scale) so heterogeneity can “shrink to homogeneity” instead of being read out of noise. This is why BCF gets near-nominal CATE interval coverage where a naïve forest overfits. - **Decision layer.** With the CATE posterior, a mailing that costs c per contact and is worth the effect τ : expected value $\mathbb{E}[\tau(x) \mid \text{data}] - c$, and — the honest version — target where $P(\tau(x) > c \mid \text{data})$ is high, not merely where the mean exceeds c .

```

import numpy as np, pymc as pm, pymc_bart as pmb, arviz as az

X = np.column_stack([X_cov, T]) # treatment as a feature (S-learner)
with pm.Model() as model:
    X_data = pm.Data("X_data", X)
    mu = pmb.BART("mu", X=X_data, Y=y, m=100)
    sigma = pm.HalfNormal("sigma", 1.0)
    pm.Normal("y_obs", mu=mu, sigma=sigma, observed=y, shape=mu.shape)
    idata = pm.sample()
# conversion (0/1) outcome: p = pm.Deterministic("p", pm.math.invlogit(mu));
# pm.Bernoulli("y_obs", p=p, observed=y, shape=mu.shape) # τ then lives on the probability scale

X1 = np.column_stack([X_cov, np.ones(len(X_cov))])
X0 = np.column_stack([X_cov, np.zeros(len(X_cov))])
with model:
    pm.set_data({"X_data": X1}); mu1 = pm.sample_posterior_predictive(
        idata, var_names=["mu"]).posterior_predictive["mu"]
    pm.set_data({"X_data": X0}); mu0 = pm.sample_posterior_predictive(
        idata, var_names=["mu"]).posterior_predictive["mu"]

cate = mu1 - mu0 # posterior CATE per customer
p_worth = (cate > 10.0).mean(("chain", "draw")) # P(uptake beats the €10 cost)
target = p_worth > 0.8

```

Decision it drives. Rank by incremental value; mail only where it *confidently* pays. Show managers the customer whose point estimate says “target” but whose interval spans zero — the reversal is the whole pitch.

2. Does the offer work better for some segments? · pathmc

Business question. Does email lift high-value customers more than bargain-hunters? Quantify per segment and check whether the gap is real.

The math. - Estimand via the do-operator: $\tau(x) = \mathbb{E}[Y \mid \text{do}(T=1), X=x] - \mathbb{E}[Y \mid \text{do}(T=0), X=x]$. In a structural causal model, $\text{do}(T=t)$ deletes T ’s structural equation and clamps its value; pathmc realizes this with `pm.do` graph surgery and forward-simulates (**g-computation**), carrying full posterior uncertainty through the DAG. - **Where heterogeneity comes from.** With a linear structural equation $Y = \beta_0 + \beta_1 T + \beta_2 X + \beta_3 (T \cdot X) + \varepsilon$ you get $\tau(x) = \beta_1 + \beta_3 x$ — **heterogeneity is exactly the moderation term β_3** . Without an interaction (or random slope), τ is constant and `cate()` collapses to `ate()`. Random slopes are hierarchical moderation: $\beta_{1,g} \sim \mathcal{N}(\mu_1, \tau_1^2)$ across segments g , partially pooled so thin segments borrow strength. - **Identification** is still §-Foundations unconfoundedness on the $T \rightarrow Y$ link; the SCM just makes the adjustment set explicit and lets you query subgroup effects as functionals of the posterior.

```

import pathmc
spec = """
spend ~ b_email*email + b_val*prior_value + trend
""" # add moderation / random slopes for heterogeneity
m = pathmc.model(spec, data=df); m.fit(draws=1000, chains=4)
m.ate("spend", "email", values=(0, 1))
m.cate("spend", "email", condition={"prior_value": high_value})
m.cate("spend", "email", condition={"prior_value": low_value})

```

Decision it drives. Segment-specific send rules, with an explicit read on whether the segments truly differ (posterior on β_3).

3. Price elasticity by region · pathmc

Business question. Regional price moves. Elasticity differs by region; price and demand are confounded (we cut prices *because* demand was soft). Estimate an elasticity that survives that.

The math. - **Constant-elasticity (log-log) demand:** $\log Q_{rt} = \alpha_r + \beta_r \log P_{rt} + \gamma^\top Z_{rt} + \varepsilon_{rt}$, where $\beta_r = \partial \log Q / \partial \log P$ is the **price elasticity** for region r . - **Random slopes / partial pooling:** $\beta_r \sim \mathcal{N}(\mu_\beta, \tau_\beta^2)$. Shrinkage is a feature here — noisy regions are pulled toward the fleet mean, and τ_β quantifies how much elasticity genuinely varies. - **Confounding is the crux.** Price is endogenous: a demand shock u_t moves both P and Q , so $\text{Cov}(\log P, \varepsilon) \neq 0$ and OLS/naïve Bayes elasticity is biased toward zero (or worse). Conditioning on the **backdoor set** Z (seasonality, competitor price, promotions, trend) closes the observable paths; $\text{lag}(\text{price})$ handles dynamic carry-over. **What partial pooling does not fix:** an *unobserved* common shock — for that you need an instrument (\$11, e.g. a cost shifter) or a natural experiment. State this assumption explicitly when presenting.

```
spec = """
demand ~ b_price*price + b_lag*lag(price)
      + b_comp*competitor_price + b_season*season + trend
"""
m = pathmc.model(spec, data=panel_df) # panel indexed by region x week; random slope on price
m.fit(); m.effects_summary()
```

Decision it drives. Region-by-region pricing, with low-confidence regions routed to a controlled price test before acting.

4. Which funnel stage to invest in · pathmc

Business question. Fixed budget across onboarding / activation / retention — which stage has the most leverage on paid conversion, respecting that effects cascade?

The math. - **Chain SCM** $X \rightarrow M \rightarrow Y$. **Effect decomposition** (Pearl’s natural effects): Total Effect = Natural Direct Effect + Natural Indirect Effect,

$$\text{NDE} = \mathbb{E}[Y(1, M(0)) - Y(0, M(0))], \quad \text{NIE} = \mathbb{E}[Y(1, M(1)) - Y(1, M(0))].$$

Under a **linear** SCM these reduce to the classic **product-of-coefficients**: for $M = aX + \dots$, $Y = bM + cX + \dots$, the indirect effect is $a \cdot b$ and the direct effect is c (the “difference method” $c = \text{TE} - ab$ agrees). pathmc computes path-specific effects by propagating posterior draws through the intervened graph, so nonlinear chains work too. - **Assumptions (stronger than a single-link analysis):** no unmeasured confounding of $X \rightarrow Y$, $M \rightarrow Y$, and $X \rightarrow M$; plus the cross-world condition that no $M \rightarrow Y$ confounder is itself affected by X . Natural effects are not identified if that last one is violated — worth flagging.

```
spec = """
engagement ~ e_ch*channel_quality + e_on*onboarding_score
activated  ~ a_eng*engagement + a_ch*channel_quality
converted  ~ c_act*activated + c_eng*engagement
"""
m = pathmc.model(spec, data=df); m.fit()
m.effect("onboarding_score -> engagement -> activated -> converted") # path-specific effect
m.prob("converted > 0", set={"onboarding_score": high})                # what-if intervention
```

Decision it drives. “€X on onboarding buys Y conversions vs Z from activation,” with uncertainty — high-leverage-but-uncertain stages become experiments, not blind bets.

5. What to control for — and what NOT to • pathmc / pymc-marketing

Business question. Which variables belong in the model? Controlling for the wrong thing silently biases everything above.

The math. - d-separation & the backdoor criterion. Z is admissible for (T, Y) if (a) no node in Z is a descendant of T , and (b) Z blocks every path between T and Y that has an arrow *into* T . Then $P(Y | do(T)) = \sum_z P(Y | T, Z=z)P(z)$ — the adjustment formula of §-Foundations. - **Colliders open paths.** For $T \rightarrow C \leftarrow Y$, conditioning on the collider C (or a descendant) *induces* dependence between T and Y — this is **selection / M-bias**. So “opened-email” (a post-treatment collider) must stay *out* of the adjustment set. Over-controlling for a **mediator** likewise removes part of the effect you’re trying to measure. - **Front-door** (fallback under unmeasured T – Y confounding, if a fully-mediating measured M exists): $P(Y | do(T)) = \sum_m P(m | T) \sum_{t'} P(Y | m, t')P(t')$. - **Sensitivity analysis** (identification is untestable, so bound the damage): quantify how strong an unobserved confounder U — via its associations with T and with Y — must be to nullify the effect. Modern tools: the **E-value**, and **partial- R^2 / robustness value** (Cinelli-Hazlett 2020), RV_q = the confounder strength (as fraction of residual variance) needed to reduce the estimate by q . The Bayesian route places a prior on U ’s strength and marginalizes.

```
m = pathmc.model(dag_spec) # DAG only – no data needed
m.adjustment_sets("email", "spend") # admissible control sets
m.collider_warnings({"opened_email"}, "email", "spend") # flags induced bias
m.is_identifiable("email", "spend")
# ... after fitting ...
m.sensitivity("spend", "email") # tipping point vs hidden confounding
```

Decision it drives. Model specs you can defend to a skeptical CMO (“here’s why we did *not* control for opens”), plus a robustness statement on hidden confounding.

6. Is spend *causing* sales? Budget allocation • pymc-marketing

Business question. Is brand search *creating* demand or catching people upper-funnel channels already warmed up? Last-click attribution over-credits it and misallocates budget.

The math. - MMM likelihood:

$$y_t = \text{baseline}_t + \sum_c \text{sat}(\text{adstock}(x_{c,t})) \beta_c + \gamma^\top Z_t + \varepsilon_t.$$

Geometric adstock (carry-over): $x_{c,t}^* = x_{c,t} + \lambda_c x_{c,t-1}^*$, $\lambda_c \in [0, 1)$. **Saturation** (diminishing returns), e.g. Michaelis-Menten / Hill: $\text{sat}(x) = \frac{\alpha x}{\kappa + x}$. These are nonlinear response transforms with priors on $(\lambda, \alpha, \kappa, \beta)$. - **Causal layer.** The DAG + backdoor criterion decides which columns are *controls* (confounders — e.g. seasonality that drives both spend and sales) and which are **mediators** that must be left out (channel $\rightarrow$ channel $\rightarrow$ sales; adjusting for them severs the very path you’re pricing). `treatment_nodes` marks the channels; the estimand for channel c is the interventional contribution $\mathbb{E}[y | do(x_c)] - \text{baseline}$, and **ROI** = contribution / spend, with a posterior. This is why a *causal* MMM $\neq$ a regression with all columns thrown in.

```
from pymc_marketing.mmm import MMM, GeometricAdstock, MichaelisMentenSaturation
dag = """
digraph { tv -> consideration; consideration -> brand_search; consideration -> sales;
          brand_search -> sales; seasonality -> sales; seasonality -> brand_search; }
"""
mmm = MMM(
    date_column="date_week", target_column="sales",
    channel_columns=["tv", "brand_search"], control_columns=["seasonality"],
    adstock=GeometricAdstock(l_max=8), saturation=MichaelisMentenSaturation(),
    outcome_node="sales", dag=dag, treatment_nodes=["tv", "brand_search"],
```

```
)
mmm.build_model(X=X, y=y); mmm.fit(X, y)
```

Decision it drives. Budget split on *incremental* contribution with credible intervals on each channel's ROI — uncertain channels get test budget, not full commitment.

7. Did the regional campaign work? • CausalPy — Synthetic Control

Business question. TV ran in a few DMAs, the rest dark, no randomized control. What would treated-DMA sales have been without it?

The math. - Setup: one treated unit, J controls (donor pool), T_0 pre-periods. Choose weights $w = (w_1, \dots, w_J)$ on the **simplex** ($w_j \geq 0$, $\sum_j w_j = 1$) so the weighted donors track the treated unit pre-treatment: $\min_w \|Y_1^{\text{pre}} - \sum_j w_j Y_j^{\text{pre}}\|$. Counterfactual $\hat{Y}_{1,t}(0) = \sum_j w_j Y_{j,t}$; effect $\hat{\tau}_t = Y_{1,t} - \hat{Y}_{1,t}(0)$ for $t > T_0$. CausalPy's WeightedSumFitter is the Bayesian version: a Dirichlet-type prior on the simplex → **posterior over the counterfactual path** → HDI on the impact. - **Assumptions:** the **convex-hull / no-extrapolation condition** (the treated unit must sit inside the donor convex hull — CausalPy warns if not; the augmented SC method relaxes it); no anticipation; no spillover from treated to donors; donors themselves untreated. Simplex weights (vs unconstrained regression) are what prevent extrapolation and negative weights.

```
import causalpy as cp
result = cp.SyntheticControl(
    df, treatment_time=launch_week,          # wide: index=week, cols=DMAs
    control_units=["dma_b", "dma_c", "dma_d", "dma_e"], treated_units=["dma_a"],
    model=cp.pymc_models.WeightedSumFitter(sample_kwargs={"target_accept": 0.95}),
)
result.summary(); result.plot()
```

Decision it drives. “Campaign drove $\sim X$ incremental sales (HDI [lo,hi])” → national-rollout go/no-go.

8. Did the loyalty rollout work? • CausalPy — Difference-in-Differences

Business question. Loyalty app launched in some stores, not others. Did revenue rise *because of it*, net of the seasonal uptick everyone saw?

The math. - 2×2 estimand:

$$\tau = (\mathbb{E}[Y \mid \text{treat, post}] - \mathbb{E}[Y \mid \text{treat, pre}]) - (\mathbb{E}[Y \mid \text{ctrl, post}] - \mathbb{E}[Y \mid \text{ctrl, pre}]),$$

which is exactly the interaction coefficient β_3 in $Y = \beta_0 + \beta_1 \text{group} + \beta_2 \text{post} + \beta_3(\text{group} \times \text{post}) + \varepsilon$. - **Key assumption — parallel trends:** absent treatment, treated and control would have moved in parallel, i.e. $\mathbb{E}[Y(0) \mid \text{treat, post}] - \mathbb{E}[Y(0) \mid \text{treat, pre}] = \mathbb{E}[Y(0) \mid \text{ctrl, post}] - \mathbb{E}[Y(0) \mid \text{ctrl, pre}]$. Untestable, but pre-period trends are the evidence; if they diverge pre-launch, use §7 instead. - **Staggered-adoption trap:** with heterogeneous timing, two-way fixed-effects DiD uses already-treated units as controls (“forbidden comparisons”) and can put **negative weights** on some effects (Goodman-Bacon decomposition), biasing the estimate under dynamic effects. Use modern estimators (Callaway-Sant’Anna, Sun-Abraham) for staggered rollouts — a caveat worth having ready.

```
result = cp.DifferenceInDifferences(
    df, formula="revenue ~ 1 + group*post",
```

```
time_variable_name="t", group_variable_name="group",
model=cp.pymc_models.LinearRegression(),
)
result.summary(); result.plot()
```

Decision it drives. A defensible “€X/store/week” for the rollout case.

9. Does the “€100 → Gold tier” perk retain people? · CausalPy — Regression Discontinuity

Business question. Cross €100 annual spend → Gold status. Does the *perk* boost retention, or do big spenders retain anyway?

The math. - **Sharp RD:** treatment is deterministic in the running variable, $T = \mathbf{1}\{X \geq c\}$. Estimand:

$$\tau_{RD} = \lim_{x \downarrow c} \mathbb{E}[Y \mid X = x] - \lim_{x \uparrow c} \mathbb{E}[Y \mid X = x].$$

- **Identification:** potential-outcome means $\mathbb{E}[Y(t) \mid X = x]$ are **continuous at c** , so units just below vs just above are comparable and any jump in Y at c is causal. This is a **LATE at the cutoff** — local to $x = c$ (exactly the margin where you’d retune the threshold). - **Estimation:** local-linear regression on each side within a bandwidth h (bias-variance trade-off; global high-order polynomials are discouraged — Gelman-Imbens). **Fuzzy RD** (probability of treatment jumps but not $0 \rightarrow 1$): $\tau = \frac{\text{jump in } Y}{\text{jump in } P(T=1)}$, an IV-flavored ratio. **Validity checks:** no manipulation of the running variable (McCrary density test), and watch for heaping at round numbers.

```
result = cp.ReggressionDiscontinuity(
    df, formula="retention ~ 1 + spend + treated",
    running_variable_name="spend", treatment_threshold=100.0,
    model=cp.pymc_models.LinearRegression(),
)
result.summary(); result.plot()
```

Decision it drives. Whether the perk earns its cost, and where to set the cutoff.

10. Did the site redesign help? (no control group) · CausalPy — Interrupted Time Series

Business question. Homepage redesigned on a known date for *all* traffic — no control group. Real lift, or normal drift?

The math. - **Model + counterfactual:** fit the pre-period series $f(t)$ (level, slope, seasonality) and extrapolate it past the intervention as the “no-redesign” counterfactual $\hat{Y}_t(0)$; effect = $Y_t - \hat{Y}_t(0)$ post-intervention. Bayesian → posterior-predictive counterfactual, so the **uncertainty band widens with horizon** (honest extrapolation). - **Identification assumption (the weakest of the CausalPy set):** absent the redesign, the pre-trend would have continued — i.e. **no concurrent shock** at the intervention date (no competing campaign, price change, or seasonality break coinciding with it). Best when the change truly hit everyone at once and the pre-period is long and stable.

```
result = cp.InterruptedTimeSeries(
    df, treatment_time=redesign_date,
    formula="conversion ~ 1 + t + month",          # trend + seasonality
)
```

```
model=cp.pymc_models.LinearRegression(),
)
result.summary(); result.plot()
```

Decision it drives. Keep or roll back the redesign.

11. Ad exposure is self-selected — the true effect • CausalPy — Instrumental Variables

Business question. People who see our ads convert more, but engaged customers also seek us out — exposure is endogenous. What’s the true effect?

The math. - **Endogeneity:** $\text{Cov}(T, \varepsilon) \neq 0$ in $Y = \beta T + \varepsilon$, so OLS is biased. An **instrument** Z must satisfy: **relevance** $\text{Cov}(Z, T) \neq 0$; **exclusion** Z affects Y only through T ; **exogeneity** $Z \perp \varepsilon$. (Good marketing instruments: randomized *encouragement* to opt in, ad-auction price shocks, geo/time assignment lotteries.) - **Wald / 2SLS estimator:** for binary Z , $\hat{\beta}_{IV} = \frac{\text{Cov}(Z, Y)}{\text{Cov}(Z, T)}$; in

general, two-stage least squares regresses T on Z (first stage) then Y on the fitted $\hat{T}$. - **What IV actually identifies — LATE** (Imbens-Angrist): under **monotonicity** (no “defiers”), IV recovers the effect **for compliers only** — the sub-population whose exposure the instrument moves — not the ATE. Report it as such. - **Bayesian IV** (what CausalPy fits): a joint model of (T, Y) with **correlated errors**, $\begin{pmatrix} \varepsilon_T \\ \varepsilon_Y \end{pmatrix} \sim \mathcal{N}(0, \Sigma)$; the off-diagonal ρ is the endogeneity, estimated rather than assumed away. **Weak instruments** (small first-stage $\text{Cov}(Z, T)$) inflate variance and bias — check first-stage strength.

```
result = cp.InstrumentalVariable(
    instruments_data=df[["ad_exposure", "encouragement"]],
    data=df[["sales", "ad_exposure"]],
    instruments_formula="ad_exposure ~ 1 + encouragement", # first stage
    formula="sales ~ 1 + ad_exposure", # structural equation
    model=cp.pymc_models.InstrumentalVariableRegression(),
)
```

Decision it drives. A believable exposure→sales number (for the *compliers*) to set budget and frequency caps, corrected for self-selection.

How this maps back to Michele’s session

- **HTE + CausalML/XAI** core → §§1–2 (pymc-bart uplift, pathmc segment effects): his questions, with per-customer uncertainty added.
- **Identification / DAG** material → §§5–6 (pathmc, pymc-marketing).
- **“Real business is quasi-experimental”** → §§7–11 (CausalPy): what MBAs recognize from their own firms.

Suggested spine: anchor on **\$1 (uplift)**; fold in a short **\$5** (“how we knew what to control for”) to bridge to his lectures; keep **\$7 or \$8** as the “when you can’t randomize” business closer.

Reference points (for your own reading, not the slides)

Rubin (potential outcomes) · Pearl (do-calculus, backdoor/front-door, mediation) · Imbens-Angrist (LATE) · Chipman-George-McCulloch (BART) · Hahn-Murray-Carvalho 2020 (BCF) · Künzel et al. 2019 (meta-learners) · Abadie et al. (synthetic control) · Goodman-Bacon / Callaway-Sant’Anna (staggered DiD) · Cinelli-Hazlett 2020 (sensitivity).